DHEERAJ KUMAR

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SUMMARY

B.Tech Mechanical Engineering student with hands-on experience in CAD modeling, sheet metal design, weldments, and mechanical assemblies using SolidWorks andAutoCAD. Experienced in finite element analysis (FEA) using ANSYS Workbench and engineering computation using MATLAB. Seeking a Mechanical Design or CAE internship to apply and further develop practical engineering skills.

SKILLS

CAD & DESIGN:	SolidWorks (Sheet Metal, Weldments, Assemblies), AutoCAD (2D Drafting), GD&T
CAE & SIMULATION:	ANSYS Workbench (Static Structural, Transient Structural, Steady-State Thermal, Transient Thermal, Eigenvalue Buckling, Mesh Sensitivity Analysis)
COMPUTATION & TOOLS:	MATLAB (Numerical Analysis, Engineering Computations), Microsoft Excel,
SOFT SKILLS:	Problem Solving, Critical Thinking, Creativity, Adaptability
PROGRAMMING:	C, C++

ACADEMIC PROJECTS

Structural Analysis & Validation of Simply Supported Beam

- **Simulation:** Conducted staticstructuralanalysisinANSYS2025 R2 using a 1 mm mesh (40 elements through thickness) to ensure high-fidelity results.
- **Validation:** Developed a MATLAB script to calculate Euler-Bernoulli deflection and stress, achieving 100% correlation with FEA data.
- **Results:** Verified Max Deflection of -1.459 mm and Max Stress of 58.56 MPa.
- ***Tools Used:*** ANSYS Workbench, MATLAB

Structural Optimization of Leisure Chair (17-Part Assembly)

- **Design:** Engineered a 17-part assembly in SolidWorks and performed structural testing under a 1000N service load.
- **Optimization:** Reduced total mass by 65% (19.9kg to 6.87kg) using Aluminium Alloy while maintaining a Factor of Safety of 1.36.
- **Simulation:** Validated stress distribution and deformation limits using ANSYS Static Structural.
- ***Tools Used:*** SolidWorks (Sheet Metal, Weldments), ANSYS Workbench

Hydraulic Cylinder Assembly (SolidWorks)

- **Modeling:** Developed a functional 3D assembly including piston, rod, barrel, and end caps with precise mechanical constraints.
- **Mechanism:** Developed a functional 3D assembly including piston, rod, barrel, and end caps with precise mechanical constraints.
- **Integration:** Modeled critical features such as seal grooves and standard fastener locations for assembly integrity.
- ***Tools Used:*** SolidWorks

EDUCATION

Bachelor of Technology (B.Tech) in Mechanical Engineering **2023 – 2027**

Delhi Skill and Entrepreneurship University, Delhi